Keystone Environmental
Knowledge-Driven Results

October 28, 2011

Mr. Remi Odense
Ministry of Environment
PO Box 9342 Stn Prov Govt
Victoria, BC V8W 9M1

Dear Mr. Odense:

Re: Response to MOE E-mail
RE: Request for Preapprovals under Protocol 6 for Jordan River sites
Project No. 9749 (105)

The following information is being provided in response to your aforementioned e-mail dated September 15, 2011 in which additional information and clarifications were requested to support the preapproval application.

The questions are presented below followed by our responses.

Comment #1:

"Toxicity Testing Preapproval

Will the results of the toxicity tests be expressed in terms which will enable comparison to the EC20 benchmark for the proposed weight of evidence evaluation?"

Response #1: Yes, each test will result in a benchmark that is similar to an EC20, that being the inhibitory concentration with effects to 25% of the test species (IC25). The test water samples will be analyzed for basic chemistry including the COPCs in question and other potentially useful and/or confounding factors. Since the tests will require a dilution series and the IC25 will be expressed in terms of the amount of test water required to achieve the IC25 (e.g. IC25 = 75% strength test water), the result can be related back to the initial COPC concentrations measured in the test water.

Comment #2:

"Weight of Evidence Preapproval

We require that preliminary weightings be given to each line of evidence in the proposed weight of evidence evaluation. These can be modified after the data has been analysed if acceptable rationale is provided."

Suite 320
4400 Dominion Street
Burnaby, British Columbia
Canada V5G 4G3

Telephone: 604 430 0671
Facsimile: 604 430 0672
info@KeystoneEnviro.com
KeystoneEnviro.com

Response #2: Attributes similar to those presented in Menzie et al. (1996)[1] were proposed in a WOE guidance document prepared for the SABCS (2010)[2]. The SABCS attributes were considered due to their clarity and ease of use, and the fact that the SABCS attributes were adapted from Menzie et al (1996). The SABCS attributes are as follows:

a. **Strength of Association** – Site-specificity and relevance of LOE to assessment endpoint; linkage based on known biological processes; similarity of effect, mechanism of action, target organ, and level of ecological organization;
b. **Strength of Association** – This is considered the most important attribute relative to the others and therefore it was recommended that it be counted twice (i.e., double weighted);
c. **Sensitivity and Specificity** – The degree to which the LOE can detect change above baseline or reference conditions; the degree to which the LOE is specific to certain stressors; the potential for confounding factors to affect interpretation;
d. **Data Quality and Study Design** – Extent to which DQOs are met; quality of data; use of standard methods; re-evaluated during the risk characterization following a review of the amount, type, and quality of data generated;
e. **Representativeness** – Spatial and temporal overlap among measurements or samples, stressors, and ecological receptors; and,
f. **Correlation/Causation/Consistency** – Ability of LOE to demonstrate effects from exposure to stressor and to correlate effects with degree of exposure; considered in greater detail during risk characterization than in problem formulation.

Each LOE was considered in relation to how well they met the assessment endpoints via the attributes listed above (i.e., protection of lotic and benthic invertebrates, and protection of resident fish populations). Each attribute was assigned numerical weights from 1 to 5, from low to high. The assigned weights to each toxicity test/LOE and rationale are presented below in **Tables 1 to 3**.

[1] Menzie, C.A., M.H. Henning, J. Cura, K. Finkelstein, J. Gentile, J. Maughan, D. Mitchell, S. Petron, B. Potocki, S. Svirsky, and P. Tyler. 1996. Special report of the Massachusetts Weight-of-Evidence Workgroup: A weight-of-evidence approach for evaluating ecological risks. Hum. Ecol. Risk Assess. 2(2):277–304.
[2] Science Advisory Board for Contaminated Site in BC. 2010. Guidance for a Weight-of-Evidence Approach in Conducting Detailed Ecological Risk Assessment in BC. Submitted to BC Ministry of Environment, Oct. 2010.

Table 1. Weights for Algae 72-h Growth Inhibition (*S. costatum*)

LOE Attribute	Attribute Scores					
	1	2	3	4	5	Rationale
a. Strength of Association		✓				Biological processes directly link the LOE to the assessment endpoint and LOE, but the specific effect, target organ, and mechanism of action evaluated are not the same.
b. Strength of Association		✓				Same as above
c. Sensitivity & Specificity				✓		LOE can detect moderate changes with a high degree of confidence, and small changes with a low degree of confidence; five of the following six factors reflects the site (*): data*, media*, species, environmental conditions*, benchmark*, and habitat type*.
d. Data Quality & Study Design					✓	Assumes that all DQOs are met and DQOs are rigorous and comprehensive; use of a standard method. *This attribute will be considered in greater detail during the risk characterization.*
e. Representativeness			✓			Spatial overlap between study area, sampling site(s) and stressors; but does not directly overlap with ecological receptors and points of potential exposure. Measurements collected when effects would be most clearly manifested, AND a single measurement is conducted, AND moderate variability in the parameter is expected over time.
f. Correlation/Causation/Consistency		✓				LOE response to stressor suggested in previous studies, but has not been definitely proven; mechanistic linkage absent. *This attribute will be considered in greater detail during the risk characterization.*
Average LOE weight:						**Average = 3.0**

Table 2. Weights for Amphipod 14-d Growth & Survival (*H. azteca*)

LOE Attribute	Attribute Scores					
	1	2	3	4	5	Rationale
a. Strength of Association		✓				Biological processes directly link the LOE to the assessment endpoint and LOE, but the specific effect, target organ, and mechanism of action evaluated are not the same.
b. Strength of Association		✓				Same as above
c. Sensitivity & Specificity				✓		LOE can detect moderate changes with a high degree of confidence, and small changes with a low degree of confidence; five of the following six factors reflects the site (*): data*, media*, species, environmental conditions*, benchmark*, and habitat type*.
d. Data Quality & Study Design					✓	Assumes that all DQOs are met and DQOs are rigorous and comprehensive; use of a standard method. *This attribute will be considered in greater detail during the risk characterization.*
e. Representativeness			✓			Spatial overlap between study area, sampling site(s) and stressors; but does not directly overlap with ecological receptors and points of potential exposure. Measurements collected when effects would be most clearly manifested, AND a single measurement is conducted, AND moderate variability in the parameter is expected over time.
f. Correlation/Causation/Consistency		✓				LOE response to stressor suggested in previous studies, but has not been definitely proven; mechanistic linkage absent. *This attribute will be considered in greater detail during the risk characterization.*
Average LOE weight:						**Average = 3.0**

Table 3. Weights for Rainbow Trout 7-d Fry Growth & Survival (*O. mykiss*)

LOE Attribute	Attribute Scores					
	1	2	3	4	5	Rationale
a. Strength of Association			✓			LOE and assessment endpoint are directly linked and the adverse effect, target organ, and mechanism of action are the same for LOE and assessment endpoint, but the levels of ecological organization differ.
b. Strength of Association			✓			Same as above
c. Sensitivity & Specificity				✓		LOE can detect moderate changes with a high degree of confidence, and small changes with a low degree of confidence; five of the following six factors reflects the site (*): data*, media*, species, environmental conditions*, benchmark*, and habitat type*.
d. Data Quality & Study Design					✓	Assumes that all DQOs are met and DQOs are rigorous and comprehensive; use of a standard method. *This attribute will be considered in greater detail during the risk characterization.*
e. Representativeness			✓			Spatial overlap between study area, sampling site(s) and stressors; but does not directly overlap with ecological receptors and points of potential exposure. Measurements collected when effects would be most clearly manifested, AND a single measurement is conducted, AND moderate variability in the parameter is expected over time.
f. Correlation/Causation/Consistency			✓			LOE response to stressor demonstrated in previous studies, but response is not correlated with magnitude of exposure; mechanistic linkage equivocal. *This attribute will be considered in greater detail during the risk characterization.*
Average LOE weight:						**Average = 3.5**

Overall, the invertebrate and fish toxicity tests are weighed higher than the algae test. The main reasons for the differences have to do with the 'strength of association' and 'causation' attributes; the proposed invertebrate and fish tests are more directly linked to the assessment endpoints than the algae test.

Comment #3:

"Food Chain Modelling Preapproval

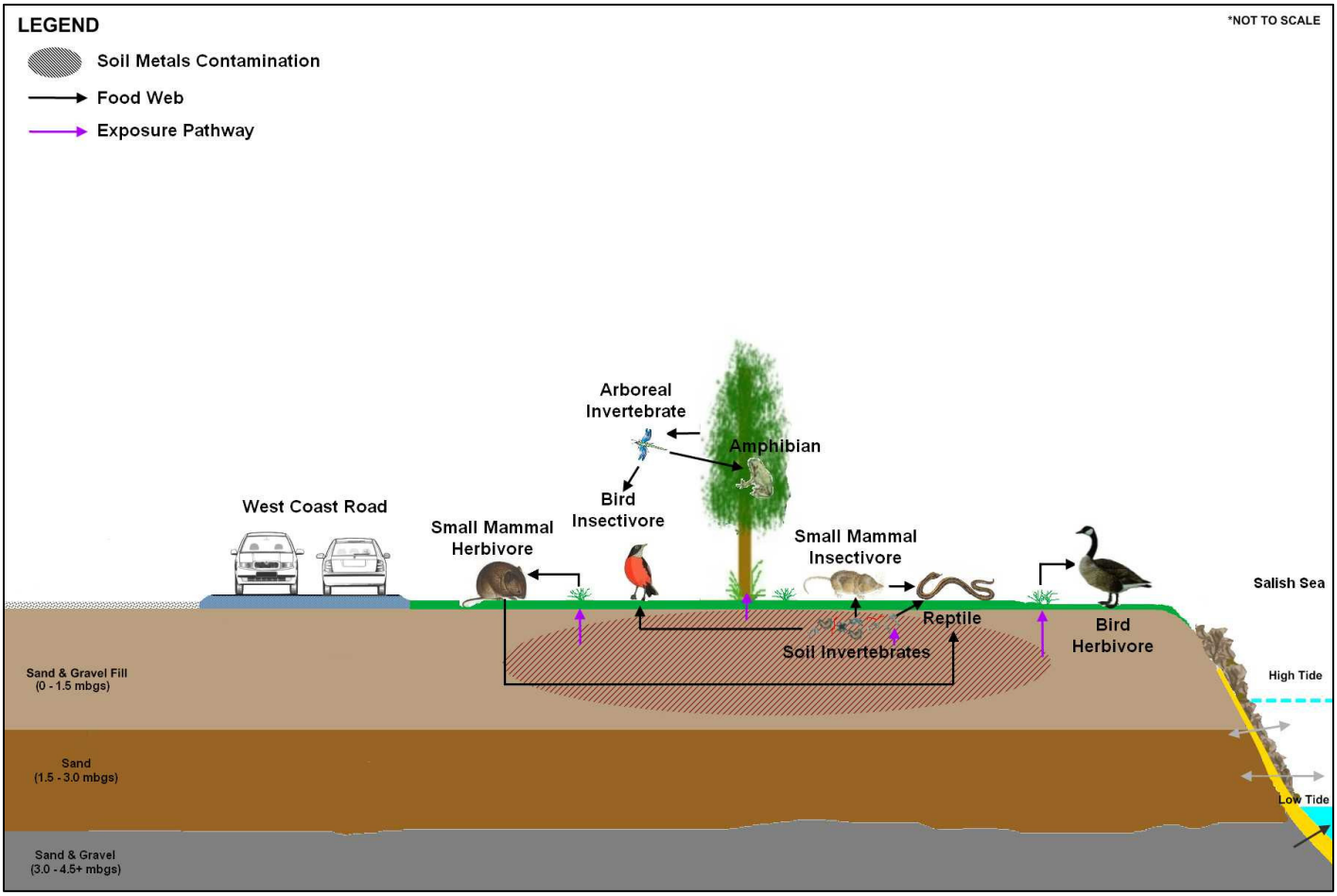
Please provide the following additional information on the proposed upland food chain model:

1. Conceptual upland food web
2. Species selected as receptors of concern
3. Details and examples of the proposed calculations for the food chain model, including references for the calculations and the sources for inputs to the model (e.g. soil ancillary parameters, wildlife factors etc.)."

Response #3: The requested items 1 to 3 of Comment #3 are provided below:

1. A conceptual upland food web is provided below in **Figure 1** depicting the receptors of concern that are proposed for the ERA.

Figure 1. Upland Conceptual Site Model



2. The species selected as receptors of concern were considered in relation to the type of COPCs present in shallow soil at the site and their potential to bioaccumulate from soil to plant and from soil to soil invertebrate tissue. Specifically the metals cadmium and zinc at the site have the highest potential to bioaccumulate; however, site COPCs are not known to substantially biomagnify up the food chain (USEPA 1999)[3]. As such, receptors were selected that have the highest potential for exposure due to direct exposure. The following is the list of receptors that are proposed along with rationale:

Receptor of Concern	**Representative Measurement Receptor**	**Rationale for Selection**
Terrestrial Plants	Mixed Grasses (Poaceae)	Primary producers; can be directly exposed to site COPCs; significant dietary component of higher trophic levels.
Soil Invertebrates	Earthworm (*Eisenia*/*Lumbricus* sp.)	Only soil invertebrate identified on-site; can be directly exposed to site COPCs; significant dietary component of higher trophic level organisms.
Litter/Arboreal Invertebrates	Beetle (Coleoptera)	Dietary component of higher trophic level organisms.
Small Mammal – ground herbivore	Townsend's Vole (*Microtus townsendii*)	Potential direct exposure to site COPCs; important to structure/function of ecosystem; and, adequate data available for this receptor.
Small Mammal – ground insectivore	Vagrant Shrew (*Sorex vagrans*)	Potential direct exposure to site COPCs; important to structure/function of ecosystem; and, adequate data available for this receptor.
Bird – ground herbivore	Canada Goose (*Branta canadensis*)	Bird most suited to site habitat and observed on-site; can be directly exposed to site COPCs; and, adequate data available for this receptor.
Bird – ground insectivore/omnivore	American Robin (*Turdus migratorius*)	Potential direct exposure to site COCs; important to structure/function of ecosystem; most sensitive bird species given dietary component of soil invertebrates; and, adequate data available for this receptor.
Amphibians	Pacific Tree Frog (*Hyla regilla*)	Adequate data available for this receptor; and serves as a surrogate receptor for the red-listed Red-legged frog that has been recorded in the Jordan River area.
Reptiles	Western terrestrial garter snake (*Thamnophis elegans*)	Potential direct exposure to site COCs; important to structure/function of ecosystem; adequate data available for this receptor; and, serves as a surrogate receptors for the red-listed Sharp-tailed snake.

[3] US EPA. 1999. Screening Level Ecological Risk Assessment Protocol for Hazardous Waste Combustion Facilities. Solid Waste and Emergency Response. EPA530-D-99-001A.

3. The following equations are proposed for the food chain modelling. An example is provided for the calculation of exposure to an American Robin (*Turdus migratorius*):

A sample calculation for the American robin's total daily oral dose of zinc is shown below. Wildlife total daily oral doses were estimated using the following model (Suter 2007)[4]:

$$E_j = (SUF)(HSF)\left[\sum_{i=1}^{m}\sum_{k=1}^{n} P_{ik}(I_i \times C_{ijk})\right]$$

Where,

E_j = Total oral dose of contaminant (j) (mg/kg/day),
SUF = Site use factor, or proportion of suitable habitat in the contaminated area (unitless),
HSF = Habitat Suitability Factor (unitless) = area contaminated / home range size,
m = Total number of ingested media (e.g, food, soil) (unitless),
I_i = Ingestion rate for medium (i) (kg/kg BW/day or L/kg BW/day),
n = Number of types of medium (i) consumed (unitless),
P_{ik} = Proportion of type (k) of medium (i) consumed (unitless),
C_{ijk} = Concentration of contaminant (j) in type (k) of medium (i) (mg/kg or mg/L).

The exposure characteristic data will be obtained from US EPA sources (e.g., 1993[5], 2003[6]). Where different receptor characteristics are made available (i.e., proportion of dietary preferences), the data best suited to the site habitat conditions will be preferably used. An example the total daily oral dose of zinc by the American robin (omnivore) is calculated below. Note that the soil invertebrate and vegetation concentrations will be based on measure field data from the site. The following are examples of the pertinent variable values used to estimate the American robin's total daily oral exposure to zinc:

Csoil = 164 mg/kg (90th Percentile or UCL 95)
$Cinvert_{soil}$ = 157 mg/kg w/w (measured, 90th Percentile or UCL 95)
$Cinvert_{foliar}$ = Cveg = 19.1 mg/kg w/w
Cveg = 19.1 mg/kg w/w (measured, 90th Percentile or UCL 95)
SUF = 1
HSF = 0.3
I = 0.9 kg/kg BW/day
Psoil = 0.02
$Pinvert_{soil}$ = 0.245
$Pinvert_{foliar}$ = 0.245
Pveg = 0.49

[4] Suter, G.W., 2007. Ecological Risk Assessment. CRC Press.
[5] US EPA. 1993. Wildlife Exposure Factors Handbook. Office of Research and Development. EPA/600/R-93/187.
[6] USEPA, 2003. Multimedia, Multipathway, and Multireceptor Risk Assessment (3MRA) Modeling System Volume II: Site-based, Regional and National Data, Section 12 – Ecological Exposure Factors.

The exposure calculations are proposed as follows:

Esoil = (1.0)(0.02) (0.9 kg/kg BW/day) (164 mg/kg) = 2.95 mg/kg/day

Eveg = (1.0)(0.49) (0.9 kg/kg BW/day) (19.1 mg/kg) = 8.42 mg/kg/day

$Einvert_{soil}$ = (1.0)(0.245) (0.9 kg/kg BW/day) (157 mg/kg) = 34.6 mg/kg/day

$Einvert_{foliar}$ = (1.0)(0.245) (0.9 kg/kg BW/day) (19.1 mg/kg) = 4.21 mg/kg/day

Ej = 0.3(Esoil + Eveg + $Einvert_{soil}$ + $Einvert_{foliar}$) = 0.3(2.95 + 8.42 + 34.6 + 4.21)
= 15.1 mg/kg/day

The total oral dose (E_j) will then be divided by an appropriate toxicity reference value (e.g. US EPA EcoSSLs bounded NOAEL) to determine the risk quotient.

I trust that this provides the information required to complete the pre-approval process.

If you have any questions, please do not hesitate to contact myself.

Sincerely,

Keystone Environmental Ltd.

Stefan Quaglia, B.Sc.
Risk Assessor